

# FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design

Amirreza Mottafegh<sup>a,†</sup>, Mincheol Park<sup>b,†</sup>, Jun-young Jo<sup>b</sup>, Jin Suk Myung<sup>a</sup>, Boyoung Y. Park<sup>b,\*</sup>, and Gwang-Noh Ahn<sup>a,\*</sup>

<sup>a</sup> Digital Chemical Research Center, Korea Research Institute of Chemical Technology, 141 Gajeong-ro, Yuseong-gu, Daejeon 34114, Republic of Korea

<sup>b</sup> Department of Biomedical and Pharmaceutical Sciences, Kyung Hee University, Seoul 02447, South Korea

<sup>†</sup> Amirreza Mottafegh and Mincheol Park contributed equally to this work.

## ABSTRACT

Translating a batch protocol into continuous flow requires chemical reasoning, engineering calculations, and equipment choices to remain consistent. We present FlowPilot, a multi-agent AI system connecting these decisions through standardized chemist intake, literature retrieval, deterministic calculation, specialist review, and inventory reconciliation. A corpus of 464 classified literature records and 2,537 handbook-derived rules supports the workflow. A common design record links stage conditions, stream assignments, equipment, and the process diagram. Matched one-shot and full-pipeline generation across three chemistry cases, five retained generator models, and three repeats yields 90 outcomes. Separate LLM judges apply 14 fixed criteria informed by deterministic checks. The mean benchmark score increases from 0.794 to 0.917, while raw judge critical flags decrease from 98 to 4. Laboratory implementation examines two connected-stage reactions: photoredox Giese addition followed by oxidation affords a sulfoxide in 97–98% NMR yield, and DPDTC-mediated amidation affords an amide in 68–86% NMR yield. Selected conditions give isolated yields of 94% and 84%, respectively. These results connect computational design consistency with experimentally demonstrated synthesis, while distinguishing proposed settings, implemented operation, and measured chemical outcomes.

## INTRODUCTION

A batch protocol describes what was done in a vessel; a continuous-flow design must also specify how materials move through a connected process. This distinction matters because the advantages of flow

chemistry arise from the interaction between chemistry and reactor operation, not from miniaturization alone.<sup>1-4</sup> Controlled heat and mass transfer, irradiation, electrochemical interfaces, and gas delivery have enabled pharmaceutical synthesis and transformations that are difficult to operate reproducibly in batch.<sup>5-10</sup> Multistep and on-demand systems further demonstrate the value of coordinating reaction stages with downstream operations.<sup>11-14</sup> Nevertheless, available hardware and a successful batch procedure do not uniquely determine a suitable flow configuration.<sup>15-16</sup> The practical challenge is to turn chemical intent into a process specification whose reaction sequence, feeds, operating conditions, and equipment remain mutually consistent.

The decisions are coupled across length scales and stages. Reactor geometry affects transport and pressure loss; heating and mixing affect the conditions experienced by the reacting stream; and an interstage addition changes both composition and the volumetric flow entering the next reactor.<sup>17-21</sup> Consequently, a residence time cannot be selected independently of reactor volume, stream flow rates, and the basis used to describe a multiphase mixture. Gas equivalents require a molar-flow calculation rather than a comparison of gas and liquid volumetric rates.<sup>22</sup> These dependencies become especially important in telescoped chemistry, where incomplete upstream conversion changes the material available to the downstream reaction. Integrated apixaban synthesis and photochemical routes to sartan intermediates illustrate the chemical and operational coordination required in practical multistep development.<sup>23-24</sup>

Digital tools already support substantial parts of this task. Computer-aided synthesis planning proposes routes,<sup>25-27</sup> process simulation evaluates specified configurations,<sup>28-29</sup> and Bayesian or active-learning strategies guide experimental optimization.<sup>30-34</sup> AI-informed robotic flow synthesis has connected planning to physical execution.<sup>35</sup> RoboChem-Flex and UV-visible auto-reactometry further demonstrate modular reaction optimization and LLM-assisted photochemical screening, respectively.<sup>36-37</sup> These advances establish a strong foundation for automated experimental development. The question addressed here concerns a complementary task: assembling an inspectable initial flow-process proposal from a reaction procedure, the chemist's objectives, and the equipment available in a particular laboratory.

LLM tools and scientific agents provide routes from unstructured descriptions to coordinated actions. ChemCrow and Coscientist connect chemical reasoning with tools and experimentation, while El Agente orchestrates computational chemistry.<sup>38-40</sup> Recent Co-Scientist and Paper2Agent studies extend collaborative scientific reasoning and tested reuse of computational methods.<sup>41-42</sup> Procedural systems are particularly relevant: ACRA maps literature procedures to executable chemical descriptions and available hardware, AutoLabs combines clarification, calculation and self-correction for liquid handling, and

PRISM introduces simulation feedback before robotic execution.<sup>43-45</sup> Together, these systems establish language-based interfaces, agent collaboration and hardware-aware validation as practical foundations for more integrated chemical workflows.

Direct precedents in continuous-flow chemistry sharpen the comparison. Pan et al.'s Chat-microreactor extracts literature-based flow information, predicts flow patterns, and provides knowledge-enhanced microreactor design guidance, including a piperacillin example.<sup>46</sup> Wang et al.'s SapoMind recommends and experimentally optimizes continuous lanolin saponification conditions.<sup>47</sup> Ruan et al.'s LLM-RDF has broader scope: its agents support synthesis development through experimental execution, kinetic analysis, optimization, purification, and reactor scale-up design.<sup>48</sup> These contributions extend beyond condition optimization to include reactor design and experimentally supported development. Their breadth shifts the comparison from whether a system uses agents or performs calculations to how it connects a reaction procedure, a process configuration and an available laboratory.

Related process-design work already links language to quantitative engineering representations. Tian et al. generate process topologies and configure parameters for executable simulations, evaluating the resulting flowsheets computationally.<sup>49</sup> The CeProAgents preprint organizes chemical process development into knowledge, concept and parameter cohorts, including flowsheet tasks and simulator-based optimization.<sup>50</sup> The CAAF preprint formalizes deterministic assertions and tests constraint conflicts in a constructed pharmaceutical flow-reactor design problem.<sup>51</sup> From the Aspuru-Guzik group, the El Agente Gráfico and La Agente Óptima preprints investigate typed execution state and agent-supervised optimization, including a physical flow-chemistry campaign in the latter.<sup>52-53</sup> The emerging landscape therefore spans chemical procedures, local hydrodynamics, process simulation and constraint validation, with different input-output representations and experimental boundaries.

FlowPilot focuses on translating reaction-level instructions into a connected, laboratory-specific flow configuration. This requires relating the order of chemical transformations to stream introduction, reagent concentrations and molar feeds, reactor volumes, stage-specific residence-time conventions, and compatible equipment assignments. FlowPilot addresses this integration task by constructing and reconciling these quantities in one inspectable process record. This differs in emphasis from predicting a local flow pattern, optimizing an established platform, or configuring a general process simulator, while overlapping with parts of each. **Table 1** summarizes the task-level comparison; **Table S36** provides the fuller source-based analysis. Neither comparison ranks the performance of the published systems.

**Table 1.** Scope of related LLM/agent systems and FlowPilot. Rows summarize the checked sources, not a performance ranking. The first six rows correspond to references 46, 48, 47, 49, 50, and 51, respectively; full comparison and evidence boundaries are in **Table S36**. \*Preprint.

| System                                  | Design task and output                                                         | Equipment and stage scope                                                                            | Reported validation                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Chat-microreactor<br>Pan et al., 2025   | Flow information, flow-pattern prediction, and microreactor guidance.          | Microreactor and multiphase-flow context; not presented as an inventory-bound multistage translator. | Flow-pattern tests and a piperacillin design example.                    |
| LLM-RDF<br>Ruan et al., 2024            | Synthesis development, kinetic analysis, optimization, and reactor design.     | Automated synthesis/analysis platform; experimental operations and scale-up.                         | Wet-lab synthesis, kinetics, photochemistry, and reactor-design studies. |
| SapoMind<br>Wang et al., 2026           | Condition recommendation and process optimization for lanolin saponification.  | An implemented continuous-flow microreactor; reaction-specific development.                          | Wet-lab product quality and process-greenness assessment.                |
| Text-to-Simulation<br>Tian et al., 2026 | Process topology and parameter generation for executable flowsheets.           | Simulator unit operations and connected flowsheets, rather than laboratory inventory allocation.     | Computational simulation convergence and design-time evaluation.         |
| CeProAgents<br>Yang et al., 2026*       | Knowledge, conceptual design, and parameter-optimization tasks.                | Hierarchical process-development agents and simulator tools.                                         | CeProBench tasks and Aspen Plus optimization.                            |
| CAAF<br>Zhang, 2026*                    | Constraint checking and controlled candidate revision.                         | Formal engineering invariants; constructed flow-reactor constraint problem.                          | Computational constraint-validation benchmarks.                          |
| FlowPilot<br>This study                 | Batch-to-flow proposal with topology, feeds, stage conditions, and provenance. | Supplied laboratory inventory; coupled multistage feeds, volumes, and equipment.                     | Matched architecture benchmark and two wet-lab connected-stage cases.    |

Evaluation must distinguish numerical consistency from chemical performance. ChemBench demonstrates that strong aggregate chemical knowledge can coexist with elementary errors and overconfidence, while uncertainty-calibrated optimization research documents invalid suggestions from direct LLM prompting within defined reaction search spaces.<sup>54-55</sup> In a flow proposal, valid formatting or agreement among agents cannot establish correct stoichiometry, compatible hardware, or adequate reaction time. We therefore ask whether a workflow that separates chemical interpretation, deterministic calculation, and specialist review produces more consistent and complete designs than one-shot

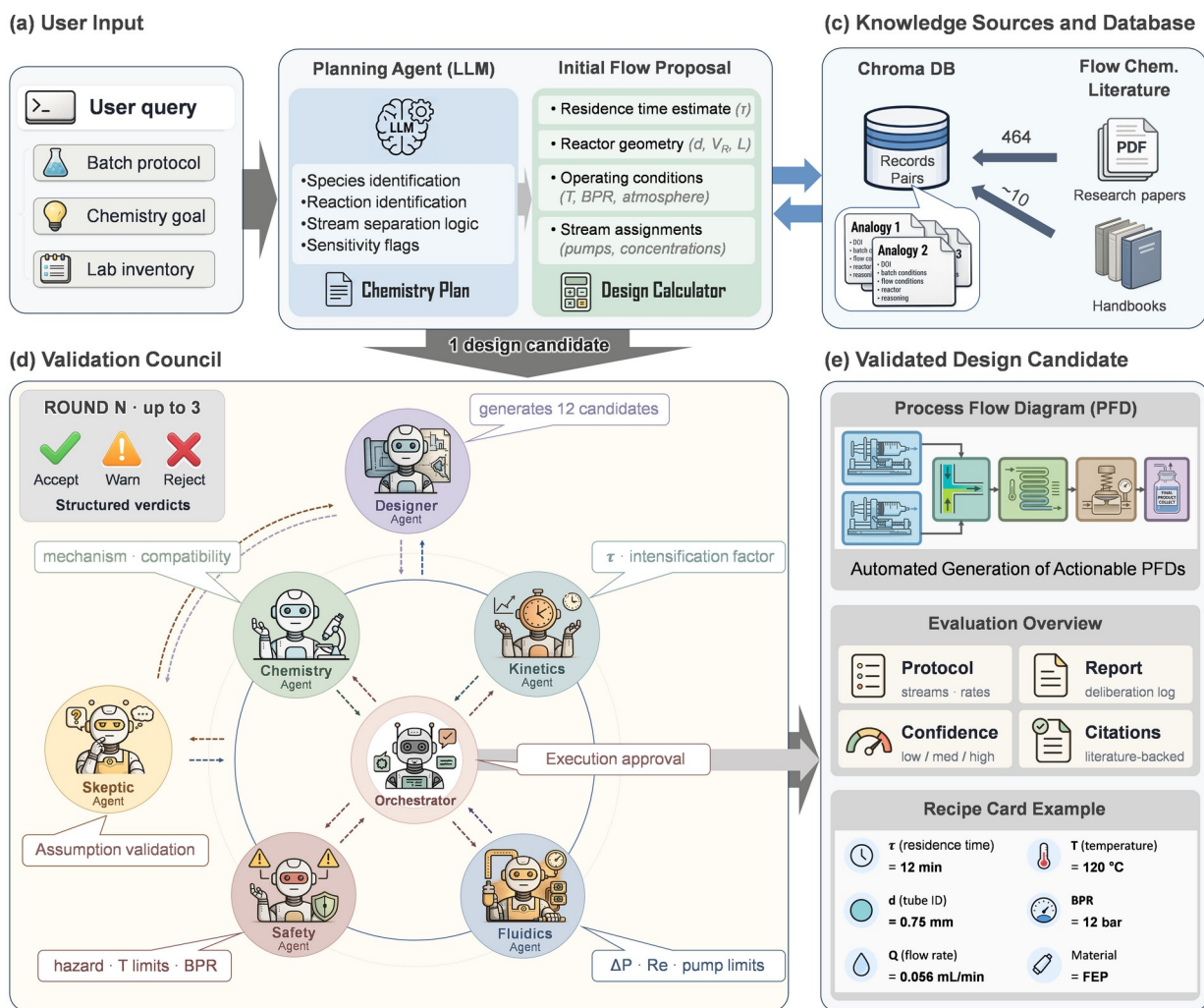
generation under the same chemistry and inventory information. The comparison tests the delivered workflows, not equal-compute performance or the isolated effect of agent count.

Here we introduce FlowPilot as a hybrid design workflow that connects standardized chemist intake, literature-grounded reasoning, quantitative engineering analysis, and inventory reconciliation. Intake records protocol facts, objectives, measured evidence, equipment limits and hypotheses separately. Chemistry interpretation and plan-aware retrieval supply relevant precedents; deterministic routines calculate dependent quantities; and a seven-agent council reviews and revises candidate operating choices. A common final record links stream tables, stage conditions, equipment assignments and the process diagram. The output is a quantitatively specified proposal for experimental implementation after laboratory review, not an automatically validated operating procedure. In particular, numerical closure cannot substitute for kinetic evidence, and a shorter residence time is a hypothesis to test rather than proof of successful intensification.

The study separates information support, computational design quality, and experimental chemical performance. Corpus and retrieval analyses evaluate the context supplied to the generator. Matched architecture comparisons use a fixed rubric, deterministic verification evidence, and separate LLM judges; module ablations and a historical isoxazole study examine configuration sensitivity.<sup>56</sup> Laboratory implementation then examines two connected-stage cases. Photoredox Giese addition followed by aerobic oxidation tests the transition from an oxygen-free stage to an oxygen-fed stage within one equipment configuration (**Figure 5**), while DPDTC-mediated amidation tests staged reagent addition, coupled flow calculations, and intermediate transfer without isolation (**Figure 6**).<sup>57-58</sup> The experiments assess implementability and product formation separately from the computational benchmark.

## RESULTS

FlowPilot organizes batch-to-flow translation around a shared process description (**Figure 1**). The chemist supplies the protocol, design objective, and laboratory inventory, then resolves standardized follow-up questions about missing chemistry, constraints, and experimental history. The confirmed answers form a stored input package for the run. The upstream module converts that context into a chemistry plan identifying species, reaction stages, stream incompatibilities, sensitivities, and possible flow-specific advantages. Retrieval supplies relevant precedents, and engineering calculations establish an initial proposal containing reactor geometry, stream assignments, residence-time assumptions, and operating conditions. Council review and equipment reconciliation then assess that proposal before it becomes the final reported design.



**Figure 1.** FlowPilot system architecture. User inputs are converted into a chemistry plan and an initial engineering proposal, supported by flow-chemistry literature and handbook knowledge. The seven-agent council generates and reviews candidate designs, records structured verdicts, and proposes bounded revisions. The selected candidate is reconciled with engineering calculations and laboratory inventory before a process-flow diagram, protocol, report, confidence assessment, and supporting citations are returned. The illustrated recipe card is schematic; computational validation does not establish experimental performance.

The downstream council supplies chemical and cross-domain review alongside deterministic numerical checks (**Figure 1d**). Rather than asking one model to produce a single final answer, the system generates a candidate pool and distributes review across domain-specialized agents. The Designer proposes candidate geometries and operating points; Chemistry Agent evaluates chemical compatibility,

photochemical constraints, and stream logic; Kinetics Agent evaluates residence time and conversion plausibility; Fluidics Agent evaluates flow regime, pressure drop, mixing, and hardware compatibility; Safety Agent evaluates pressure, thermal, material, and handling risks; the Skeptic audits assumptions and internal consistency; and the Orchestrator selects the final candidate. This structure produces not only a selected design but also a record of rejected alternatives, revised candidates, disqualified candidates, and the rationale for each decision.

For a reconciled result, the output is a structured design package containing stage and stream parameters, equipment assignments, a process diagram, supporting analogies, and the review record. Numerical views and the diagram refer to the same final design contract, while unresolved requirements remain visible. This common record connects a chemical proposal to specific equipment and makes subsequent experimental feedback attributable to the conditions actually considered. The authority boundaries, input contracts, and provenance mechanisms are described in **Figures S1-S2** and **Tables S1-S3**; computational closure does not itself establish reaction performance.

### ***Literature and rule-base coverage***

The literature dataset covers a broad range of flow-chemistry applications (**Figure 2**). The curated export contains 464 classified literature records spanning thermal synthesis, photoredox catalysis, heterogeneous photocatalysis, cross-coupling, oxidation/reduction, and other reaction classes. Reactor types range from capillary/coil and microreactor/chip systems to photoreactors and packed beds, with fluoropolymer tubing prominent among the reported materials. Explicit inlet-stream counts are available for 154 processes; two- and three-stream configurations account for 91 and 30 of these records, respectively. These annotations supply precedents rather than prescribing a default number of streams. The diversity supports retrieval across photochemical and non-photochemical processes with different materials and operating regimes.



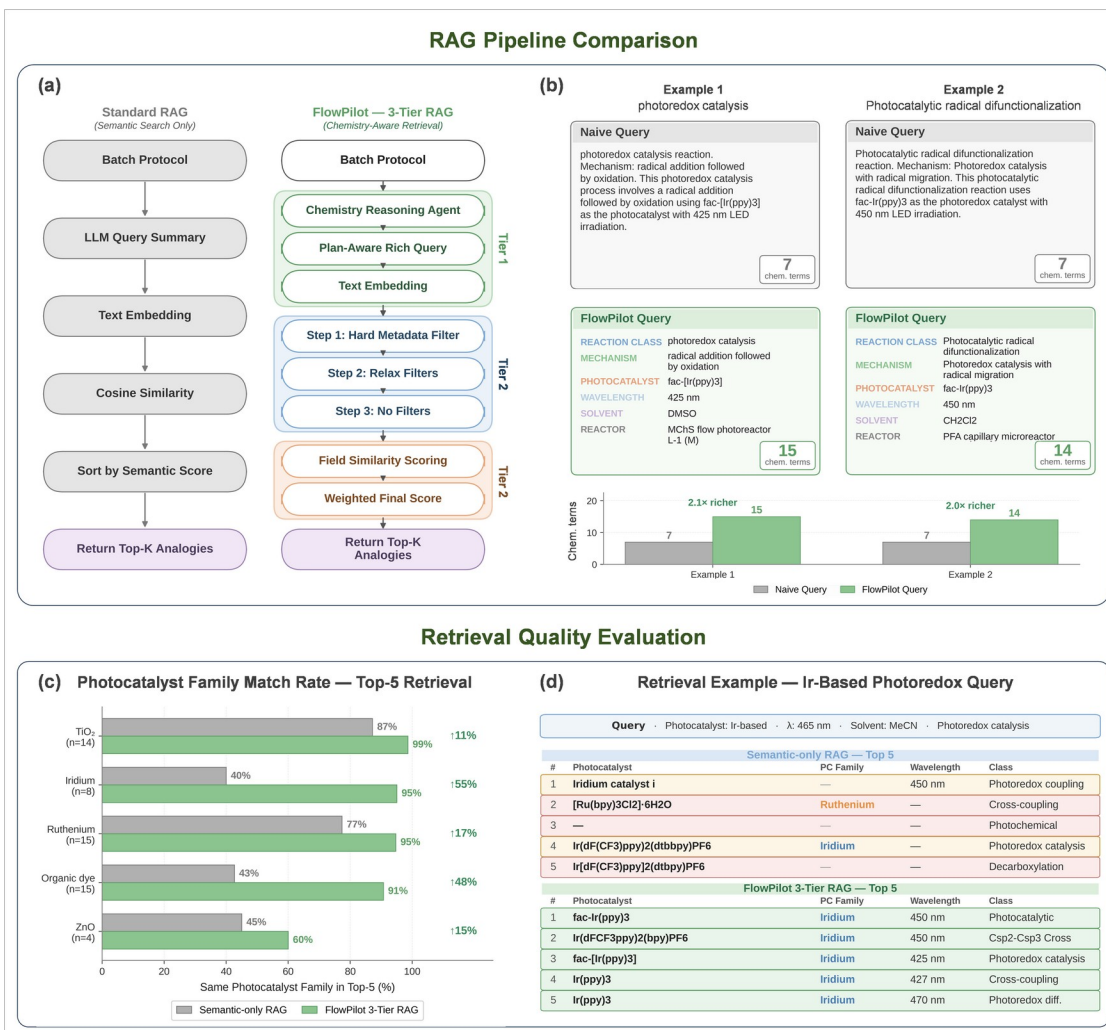
**Figure 2.** Literature corpus and engineering rule-base coverage. (a-d) Reaction classes, reactor types, reported materials, and inlet-stream configurations in the curated flow-chemistry corpus; explicit inlet-stream counts are available for 154 records. (e) Category and severity distribution within the 2,537-rule engineering knowledge base. (f) Recorded associations between rule categories and chemistry classes. Full categorical exports and field-specific availability are reported in **Figures S3-S4** and **Tables S4-S5**; associations indicate knowledge coverage rather than independent validation of every rule.

The engineering knowledge base contained 2,537 structured rules spanning reactor design, heat transfer, catalyst behavior, residence time, photochemistry, materials, scale-up, pressure, mixing, and safety (**Figure 2e-f**). Coverage was not uniform across chemistry classes: heat-transfer rules were densely associated with thermal synthesis and cross-coupling, whereas photochemistry rules concentrated in photoredox and photocycloaddition classes. This structured rule landscape supplies domain context for

evaluating model-generated designs alongside deterministic checks, rather than relying only on model priors. The frozen corpus composition and computed rule-base coverage are reported in **Figures S3-S4** and **Tables S4-S5**.

### ***Plan-aware retrieval improves analogy quality***

Coverage alone does not show that a relevant precedent will be selected. We therefore evaluated the retrieval system against a semantic-search baseline (**Figure 3**). The baseline embeds the batch protocol or query summary and ranks records by cosine similarity. FlowPilot enriches the query using the chemistry plan, including reaction class, mechanism, photocatalyst family, solvent, and wavelength. Filtering starts with paired records under mechanism and phase constraints, relaxes filters when fewer than three matches remain, and expands to all records if no paired match remains. A field-aware score then reranks the candidates. **Figure 3a** summarizes the workflow; **Figure S5** and **Table S6** specify its tier transitions and scoring weights.



**Figure 3.** Plan-aware retrieval. (a) Three-tier filtering strategy: hard metadata filter, relaxed metadata filter, and pure semantic fallback. (b) Query enrichment expands chemically specific search terms approximately two-fold relative to a naive semantic baseline. (c) Top-5 photocatalyst-family matching rates by family (iridium, organic dye, ruthenium, TiO<sub>2</sub>, ZnO) for semantic baseline vs FlowPilot reranking. (d) Representative reranking outcome showing closer wavelength and photocatalyst-family alignment.

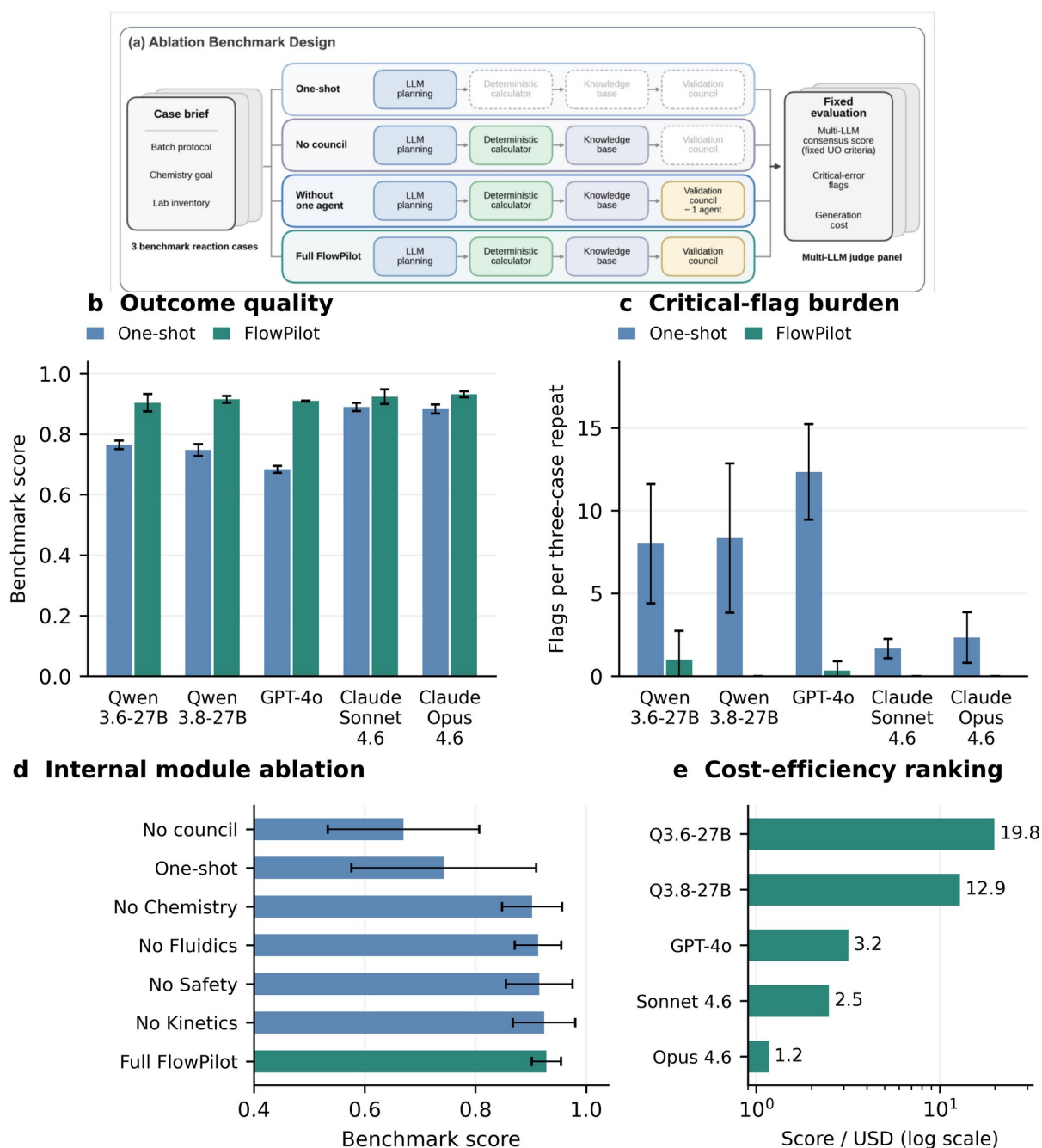
The enriched query representation approximately doubled the number of chemically specific search terms in the two examples shown in **Figure 3b**. A photoredox-catalysis query increased from 7 naive terms to 15 FlowPilot terms, and a photocatalytic radical difunctionalization query increased from 7 to 14 terms. In the retrieval-pair benchmark, 1,600 query–result pairs were evaluated. FlowPilot reranked 25.1% of

these 1,600 pairs relative to the pure semantic baseline, indicating that field-aware scoring changed the ordering of the candidate analogies presented to the design engine.

The strongest retrieval improvement was observed for photocatalyst-family matching. For iridium systems, top-five family matching increased from 40.0% under semantic search to 95.0% after reranking. Organic-dye systems improved from 42.7% to 90.7%, ruthenium systems from 77.3% to 94.7%, TiO<sub>2</sub> systems from 87.1% to 98.6%, and ZnO systems from 45.0% to 60.0% (**Figure 3c**). The demonstration in **Figure 3d** illustrates how field-aware ranking promotes records with a matching photocatalyst family and relevant wavelength context. These results measure metadata alignment, not the accuracy of a transferred kinetic estimate or the eventual reaction yield. Retrieval tiers, scoring weights, benchmark data, and source-exclusion controls are detailed in **Figures S5-S6** and **Tables S6-S7**.

### ***Ablation and architecture comparison***

Improved retrieval does not by itself establish that the resulting flow design is consistent. We therefore compared one-shot generation with the full FlowPilot pipeline while holding the chemistry, inventory, and requested output contract constant (**Figure 4a**). The three cases were CuAAC, photochemical oxidation, and hydrogenolysis, with three generation repeats per model-architecture-case combination. Separate Qwen, OpenAI, and Claude judge calls evaluated normalized final designs against 14 fixed criteria, using the same source context and deterministic verification evidence. Criteria address chemical fidelity, stream and reactor consistency, equipment feasibility, operating procedure, and uncertainty. Applicable integer ratings from 0 to 4 were averaged with equal criterion and judge weights and normalized to a score from 0 to 1. Critical flags were recorded separately. **Tables S8-S10** provide the scoring anchors, criterion definitions, and module-screen summaries.



**Figure 4. Architecture comparison and cost efficiency.** (a) Shared-input generation and fixed evaluation. (b) Benchmark score and (c) raw judge flags for five models (Tables S11–S12): mean  $\pm$  sample SD over three repeats, each containing three chemistries; flags are summed within each repeat. (d) Selected Qwen3.8-27B ablations: mean and between-chemistry SD for one generation per cell (all 15 conditions: Table S10). (e) Mean FlowPilot score divided by mean generation cost over nine outcomes per model, at recorded prices. Flags are judge findings, not independently confirmed physical errors; costs exclude judging and deployment.

Across the five-model cohort in **Table S11**, full-pipeline mean scores ranged from 0.904 to 0.932, compared with 0.684 to 0.890 for matched one-shot generation (**Figure 4b**). The mean gain was largest for GPT-4o (+0.226), followed by Qwen3.8-27B (+0.168) and Qwen3.6-27B (+0.139); the Claude gains were smaller (+0.034 and +0.049). The judges recorded 98 critical flags across the 45 one-shot outcomes and 4 across the 45 FlowPilot outcomes; 18 and 43 outcomes, respectively, had no critical flag (**Figure 4c**; **Table S12**). The score and flag summaries therefore point in the same direction within this cohort: improved completeness and fewer flagged inconsistencies. They do not establish universal model superiority or measured reaction success. **Figure 4d** provides a complementary internal comparison, with council removal causing the largest score decrease among the configurations shown.

The aggregate comparison is complemented by chemistry-specific scores (**Figure S7**), criterion-level score differences (**Figure S8**), and the location of critical flags within individual runs (**Figure S9**). **Figure S10** reports tokens, estimated generation costs, and observed runtime. The higher Qwen score-per-generation-cost values in **Figure 4e** therefore reflect the recorded token-price schedule, not a measurement of total deployment cost. **Tables S13-S14** define numerical closure and equipment checks; **Tables S15-S16** specify feedback and replay records. **Figure S11** and **Table S17** distinguish user-composed input from system-assembled prompts and returned artifacts for one matched run. This supports the practical input-handling argument without implying that FlowPilot uses fewer total model tokens. The legacy heuristic QA values in that worked example are not the fixed-criteria scores used in **Figure 4**.

The earlier council model-matrix and candidate-budget experiment is reported separately in **Figure S12** and **Tables S18-S21**. In the five-repeat comparison, the normalized radar-area score changed from 0.31 +/- 0.03 before council review to 0.48 +/- 0.23 afterward. This engineering-profile measure is not the multi-LLM benchmark score in **Figure 4**. The ESI defines Need revision, New versions, Combined pool, and Disqualified, and reports runtime, token use, geometry, and radar normalization. **Figure S13** and **Table S22** distinguish the final structured FlowPilot topology from its contemporaneous generated narrative; the complete one-shot response is retained alongside this worked example.

The ESI also shows how the chemist supplies and inspects the information behind a design. **Figures S14-S15** and **Table S23** document protocol entry and fixed-ID follow-up questions. **Figure S16** and **Table S24** show inventory import, normalization, and equipment warnings; **Figure S17** and **Table S25** show stage-resolved topology and stream parameters from a reopened run. Concrete failures are retained in **Table S26**, and **Figure S18** and **Table S27** expose actual council disagreement and selection records. **Figure S19** and **Table S28** place these views in the complete GUI and saved-run context. These

operational records connect the computational benchmark to an inspectable workflow. The laboratory cases below examine how selected proposals translate into connected reactions and measured product yields.

### ***Case study 1: photoredox Giese addition followed by aerobic oxidation***

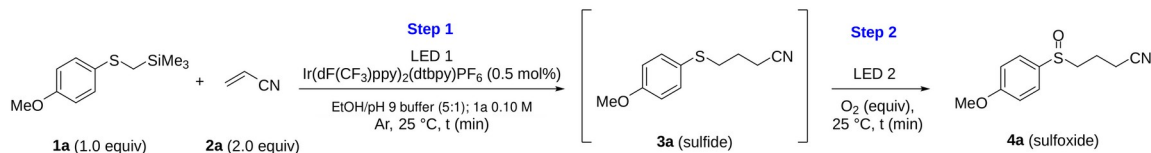
The first connected-process case couples photoredox Giese addition with oxidation of the resulting sulfide to a sulfoxide (**Figure 5**). Photoinduced single-electron transfer generates an alpha-thiomethyl radical from an alpha-silyl sulfide; addition to acrylonitrile constructs the C–C bond, and subsequent oxygen-dependent photochemistry changes the sulfur oxidation state.<sup>57</sup> This sequence requires the oxygen-free first stage and oxidizing second stage to remain distinct within a connected process. The supplied design input specifies 4 h under argon followed by 6 h in air. These input durations are retained separately from the published batch comparison reproduced by KHU, which lists 180 and 360 min. Neither is treated as a kinetic law or converted using a universal intensification factor.

The revised proposal introduces pure oxygen only between the stages and binds each coil to a compatible irradiation module. KHU implemented a 2 or 5 mL PFA coil in the Vapourtec UV-150 module followed by a separately illuminated 20 mL FEP coil, both at  $25 \pm 1$  °C. A 2.0 mL sample loop introduced a solution containing substrate 1a (0.100 M), acrylonitrile (0.200 M), and iridium photocatalyst (0.500 mM) into a carrier stream at 0.020 mL min<sup>-1</sup>. Stage 1 therefore has a nominal liquid residence time of 100 or 250 min. The oxygen-controller setting was 0.090 mL min<sup>-1</sup>. On the proposal's explicit STP basis (273.15 K, 1 atm), this corresponds to 0.00402 mmol min<sup>-1</sup> of oxygen and 2.01 equivalents relative to the nominal substrate molar flow. The laboratory MFC reference conditions require confirmation before this conversion is taken as a calibrated experimental value. The Stage 2 index,  $V/(Q_{\text{liquid}} + Q_{\text{gas,STP}})$ , is 181.82 min. It combines reference-state gas volume with liquid flow and is not the residence time of either phase at operating pressure.

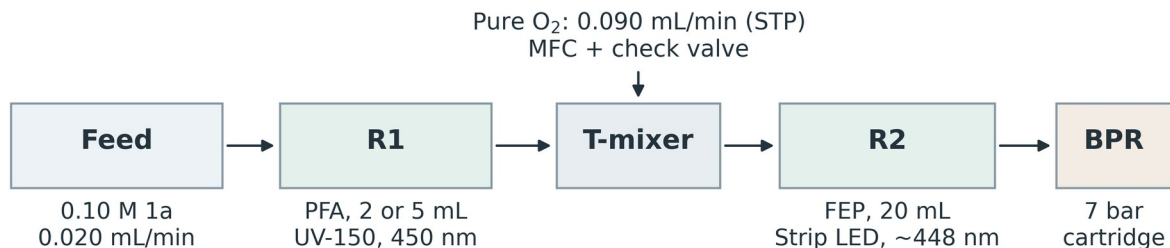
An earlier high-gas-flow trial showed upstream gas entry, prompting the move from air-based delivery to pure oxygen and a lower total gas flow. This change reduces the carrier-gas burden at a specified oxygen molar feed; it does not by itself establish hydraulic protection. The proposal includes a gas-line check valve, and the KHU photograph identifies a check valve in the assembled apparatus. Exact placement and startup-pressure behavior remain to be documented. KHU used a 7 bar cartridge BPR but recorded system pressures of 5.1–5.6 bar; the hardware rating and measured pressure are reported separately. The 2 mL first-stage configuration, grouped as Sets 1/3 in the supplied record, afforded sulfoxide 4a in 97% NMR yield and 94% isolated yield. Increasing the first-stage volume to 5 mL (Set 2) gave 98% NMR yield.

Without replicate statistics, the one-percentage-point difference does not establish a benefit from the longer first stage. **Figure S20** and **Tables S29–S31** retain the design inputs and proposed settings; ESI Section 12, **Figures S23–S25** and **Tables S37–S39** document irradiation, implementation, and characterization. These measurements demonstrate productive telescoping, not a measured residence-time distribution or an independently optimized operating point.

### a Connected chemical sequence



### b FlowPilot proposal: connected reactor train



Ar-prepared feed; O<sub>2</sub> enters only after R1. Both stages: 25 °C.

### c Reported operation and nominal stage times

| Response set(s) | R1 / R2 (mL) | R1 time (min) | R2 index* (min) | Recorded pressure (bar) |
|-----------------|--------------|---------------|-----------------|-------------------------|
| 1 and 3         | 2 / 20       | 100           | 181.8           | 5.1–5.4                 |
| 2               | 5 / 20       | 250           | 181.8           | 5.2–5.6                 |

### d Measured sulfoxide 4a yields

| Response set(s) | NMR yield (%) | Isolated yield (%) |
|-----------------|---------------|--------------------|
| 1 and 3         | 97            | 94                 |
| 2               | 98            | Not reported       |

\*R2 index = 20/(0.020 + 0.090), not operating-pressure residence time.

Shared-set entries are reported as supplied; experimental repeat counts were not provided.

**Figure 5.** Giese addition/oxidation: proposed configuration and measured outcomes. (a) Chemical sequence. (b) Simplified FlowPilot proposal (full topologies: **Figure S20**). (c) Implemented volumes and recorded pressures. R1 time is  $V/Q_{\text{liquid}}$ ; R2 uses the nominal inlet/STP index  $V/(Q_{\text{liquid}} + Q_{\text{gas,STP}})$ , not operating-pressure residence time. MFC reference conditions require confirmation. The 7 bar cartridge rating is distinct from recorded pressure. (d) NMR and isolated yields supplied by KHU. Sets 1/3 share one reported entry, not documented independent repeats. Actual sample-loop operation, photographs, procedures, and analytical data are in ESI Section 12.

### ***Case study 2: telescoped DPDTC-mediated amidation***

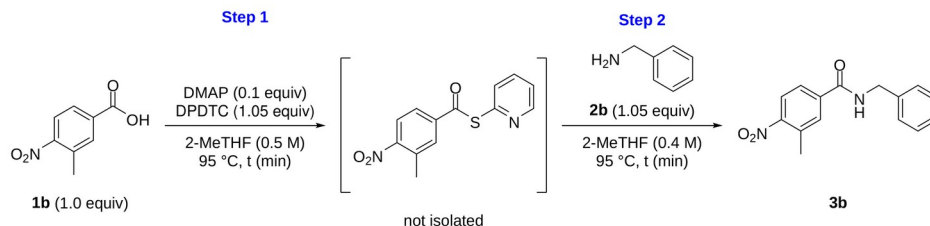
The second case is a thermal two-stage amidation of 3-methyl-4-nitrobenzoic acid with benzylamine (**Figure 6**). DPDTC and DMAP first generate a 2-pyridyl thioester; downstream aminolysis forms the amide C(O)–N bond while displacing the sulfur-containing leaving group.<sup>58</sup> The input specifies 30 min at 95 °C for each batch stage, separated by cooling and amine addition. In flow, the thioester is neither isolated nor assumed to form quantitatively. Stage 1 conversion limits the intermediate entering Stage 2, making final amide yield a response of the complete sequence rather than of either reactor alone.

KHU implemented the proposed separate-feed arrangement using two channels of the Vapourtec E-series system. Feed A contained acid (0.500 M), DPDTC (0.525 M), and DMAP (0.0500 M) in 2-MeTHF; Feed B contained benzylamine (2.10 M). In Set 1,  $Q_A = 0.160$  and  $Q_B = 0.0400$  mL min<sup>−1</sup> correspond to nominal acid and amine molar flows of 0.0800 and 0.0840 mmol min<sup>−1</sup>, preserving 1.05 equivalents of amine. Two 5 mL ETFE coils give nominal stage times of 31.25 and 25.00 min because the second reactor receives both feeds. The reported implementation used a 2.0 mL sample loop for Feed A, direct interstage addition, and a 95 °C water bath for both coils. A 7 bar cartridge BPR was fitted; the recorded pressure for Set 1 was 5.2–5.3 bar. Thus hardware selection, nominal stage times, and measured pressure remain distinct parts of the process record.

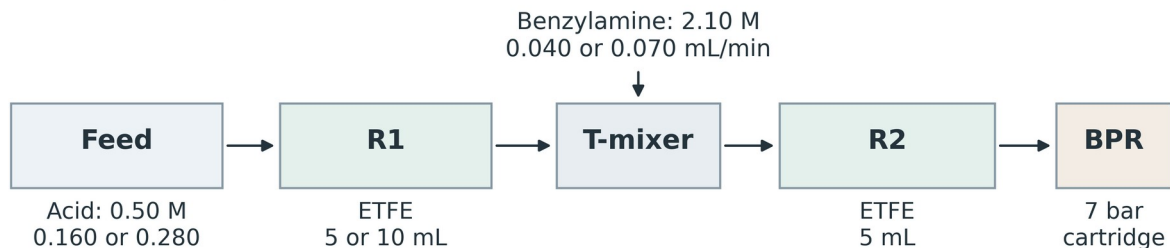
Set 1 afforded amide 3b in 86% NMR yield and 84% isolated yield. Sets 2/3 share a second reported operating point: a 10 mL first reactor and 5 mL second reactor with feeds of 0.280 and 0.0700 mL min<sup>−1</sup>, giving nominal stage times of 35.71 and 14.29 min and 68% NMR yield. The nominal combined stage time decreases from 56.25 to 50.00 min while the substrate molar feed rises from 0.0800 to 0.140 mmol min<sup>−1</sup>. The lower final yield despite a longer first stage is consistent with a coupled yield–throughput trade-off, but cannot identify the limiting step because both flow and reactor configuration changed and intermediate conversion was not measured. **Figure S21** and **Tables S32–S34** retain the response sets and design calculations. **Figures S26–S27** provide the supplied experimental scheme and apparatus photograph; **Tables S38–S39** reconcile implementation and molar feeds. The measurements support intermediate transfer without isolation while showing why numerical closure alone does not determine the best chemical operating point.

The alpha-bromination example from the original preprint<sup>59</sup> is retained unchanged as **Figure S22** in ESI Section 9 for historical comparison. It is distinct from the current amidation case and is not presented as a new laboratory validation.

### a Connected chemical sequence



### b FlowPilot proposal: connected reactor train



Flows: mL/min. Both stages: 95 °C; no intermediate isolation.

### c Reported operation and nominal stage times

| Response set(s) | $Q_A / Q_B$<br>(mL/min) | R1 / R2<br>(mL) | R1 / R2<br>time (min) | Recorded<br>pressure (bar) |
|-----------------|-------------------------|-----------------|-----------------------|----------------------------|
| 1               | 0.160 / 0.040           | 5 / 5           | 31.25 / 25.00         | 5.2-5.3                    |
| 2 and 3         | 0.280 / 0.070           | 10 / 5          | 35.71 / 14.29         | 5.5-5.8                    |

### d Measured amide 3b yields

| Response set(s) | NMR yield (%) | Isolated yield (%) |
|-----------------|---------------|--------------------|
| 1               | 86            | 84                 |
| 2 and 3         | 68            | Not reported       |

$R1 \text{ time} = V_1/Q_A$ ;  $R2 \text{ time} = V_2/(Q_A + Q_B)$ .

Shared-set entries are reported as supplied; experimental repeat counts were not provided.

**Figure 6.** DPDTC-mediated telescoped amidation: proposal and laboratory implementation. (a) Acid activation to a 2-pyridyl thioester followed by benzylamine addition and amide formation. (b) Simplified proposed connected train; complete archived topologies are in **Figure S21**. (c) Implemented flow rates, reactor volumes, nominal liquid residence times, and KHU-reported system pressures. R1 uses  $V_1/Q_A$ , whereas R2 uses  $V_2/(Q_A + Q_B)$ . Both stages were held at 95 °C; the 7 bar cartridge rating is distinct from recorded pressure. (d) Reported NMR and isolated yields. Sets 2/3 share one experimental entry; independent replicate statistics were not supplied. Sample-loop operation, detailed preparation, photographs, and analytical records are in ESI Section 12.

## DISCUSSION

FlowPilot addresses batch-to-flow translation as a connected-process design task. Chemistry interpretation, literature retrieval and specialist review determine what should be proposed, while deterministic calculations reconcile dependent quantities and equipment assignments. Their contributions are evaluated at different levels: **Figure 3** measures retrieval alignment, **Figure 4** compares delivered designs, and the component ablations examine the role of review. A shared process record connects these steps to the stream table and topology, making the relationship between chemical assumptions and engineering conditions explicit.

The matched architecture comparison shows higher mean scores with FlowPilot for each of the five retained generators, with the improvement depending on the model (**Figure 4b**). In the internal Qwen3.8-27B screen, removing the council reduced the mean score from 0.928 to 0.670 across the three cases (**Figure 4d**; **Table S10**). Individual specialist removals had smaller effects. This single-generation-per-cell screen is a descriptive component comparison, not a statistical demonstration that every specialist is necessary. The historical model-matrix and budget experiments further show that broader candidate exploration does not guarantee better outcomes (**Figure S12**; **Tables S18-S21**). Together, these results support coordinated review while showing why configuration matters.

This emphasis complements related advances in microreactor design, experimental synthesis development, process simulation, and deterministic constraint validation (**Table 1**).<sup>46-51</sup> FlowPilot links the reaction sequence to explicit feed compositions, molar flows, stage calculations, and available equipment in one record. Its focus on coupled stream, stage, and equipment consistency also complements process-optimization and materials-development agents.<sup>60,61</sup> **Table S36** describes these differences in task and output; the one-shot benchmark is an architecture comparison within this study, not a head-to-head evaluation of the published systems.

The archived examples show why record-level inspection remains important alongside aggregate scores. **Table S26** includes conflicting one-shot residence-time bookkeeping and a FlowPilot catalyst-preparation inconsistency, while **Table S22** documents disagreement between structured conditions and an accompanying narrative. These examples motivate a common final record and traceable revision decisions. Retaining the inputs, calculations, equipment selections and council exchanges makes both design conflicts and evaluator errors available for examination rather than hiding them behind a single score.

The laboratory cases connect this computational assessment to chemical outcomes. Giese addition/oxidation demonstrates a productive sequence with oxygen introduced downstream, while the closely similar yields at two first-stage volumes do not establish an optimum. DPDTC amidation demonstrates intermediate transfer without isolation, but its lower yield at higher feed rate shows that a numerically closed proposal need not be the chemically preferred operating point. Both were implemented using sample-loop injections; extended steady-state operation and repeatability are separate questions. The feedback interface can incorporate measured yields together with the actual settings and deviations, rather than treating a proposed design as experimental evidence. Integration with Bayesian optimization<sup>30-34</sup> and language-guided priors<sup>62</sup> offers a route to test chemist hypotheses against such accumulated records.

## CONCLUSION

FlowPilot connects a batch protocol and chemist-supplied constraints to a traceable first flow-process proposal. Standardized intake establishes the design context, chemistry-aware retrieval supplies precedents, deterministic calculation links dependent parameters, and specialist review assesses candidate operating choices. Equipment reconciliation and a common final record connect the selected conditions to the process diagram and supporting report.

For the five-model, three-chemistry cohort, the mean fixed-criteria score increased from 0.794 for one-shot generation to 0.917 for FlowPilot, with fewer raw judge critical flags. Module ablations support a contribution from council review, while historical model-matrix and budget studies show configuration sensitivity. Laboratory implementation produced a sulfoxide in 94% isolated yield and an amide in 84% isolated yield under selected connected-stage conditions. The different amidation outcomes at two operating points reinforce the need to evaluate chemical performance separately from computational consistency. Together, these findings support a traceable workflow from reaction-level input to an experimentally reviewable flow configuration within the scope of the cases examined.

For the chemist, the intended benefit is a shorter path from an initial procedure to an inspectable proposal: enter the protocol, resolve the standardized questions, select an inventory profile, and review the linked design records. Measurements can then be returned with the actual conditions to inform a subsequent design cycle. This connects proposal generation with experimental development while preserving the distinction between what was assumed, calculated, proposed, and observed.

More broadly, the hybrid pattern demonstrated here, LLM-driven interpretation coupled with deterministic computation and multi-agent cross-domain audit, is not specific to flow chemistry. Any

engineering design problem that is currently solved by a small team of complementary specialists iterating across coupled domains is a candidate for the same architectural approach. The modular design of FlowPilot is intended to make such extensions practical.

## METHODS

A literature corpus containing 464 classified flow-chemistry records was annotated with available reaction, reactor, material, operating-condition, yield, and stream-count metadata. Field completeness was reported separately (**Table S4**). A complementary handbook-derived engineering knowledge base<sup>3-4,21</sup> contained 2,537 rules categorized by topic and severity. The rule-base analysis describes coverage and recorded associations, while the deterministic calculator implements the numerical checks used during design.

The plan-aware retrieval pipeline first generates a chemistry plan from the user input and then constructs an expanded query containing reaction class, mechanism, photocatalyst family, and sensitivity flags. Retrieved candidates are reranked by a field-aware score that rewards mechanistic relevance over superficial semantic similarity.

The translation workflow comprises standardized intake, chemistry interpretation, retrieval, engineering calculation, candidate generation, council review, inventory realization, and topology rendering. LLMs propose chemical interpretations and independent design choices; deterministic routines recalculate dependent quantities and check their consistency. Chemist objectives and available evidence guide residence-time and throughput choices. Stage parameters, stream assignments, and topology are reconciled before final publication. Major development milestones and dated evidence are summarized in **Table S35** (ESI Section 10). Later scientific-policy and backflow extensions are separated from the frozen software used for the earlier architecture benchmarks. **Table 1** and **Table S36** compare six directly related systems with FlowPilot by task, output, equipment context, and reported validation; this literature-based comparison is not an additional performance benchmark.

The council reviews a candidate pool under bounded domain authority. Each of the seven agents (Designer, Chemistry Agent, Kinetics Agent, Fluidics Agent, Safety Agent, Skeptic, Orchestrator) may only modify fields within its own domain. Revised candidates are recomputed deterministically before final scoring.

The matched architecture comparison used the same frozen case input and final-output contract for one-shot generation and FlowPilot. Five retained generator models, two architectures, three chemistry cases,

and three generation repeats yielded 90 outcomes in the ESI cohort. Three judge families received identity-masked final designs, source context, deterministic verification evidence, and the fixed rubric in **Tables S8-S9**. Each applicable criterion was rated from 0 to 4; ratings were averaged across judges and criteria, then divided by four. Gas bookkeeping and multistage closure were omitted only when inapplicable. Per-model error bars summarize the sample SD of three repeat-level means, each averaging the three cases. Critical flags were recorded separately. Exact inputs, model records, applicability rules, and archived evaluation limitations are given in ESI Section 3.

The internal module screen fixed the generator to Qwen3.8-27B and generated each condition-case combination once; its SD therefore measures variation between chemistries, not repeatability. **Figure 4d** shows selected conditions and **Table S10** provides the displayed module summaries. The separate historical model-matrix experiment crossed four upstream and four council endpoints, and the candidate-budget study tested  $B = 1, 6, 12$ , and 24 with five repeats. Their validation dispositions, numerical data, and metric definitions are reported in ESI Section 6.1 and **Tables S18-S21**. These studies are kept distinct from the independent-judge outcome benchmark.

Laboratory experiments were performed by KHU using a Vapourtec E-series system. The photochemical sequence used PFA and FEP coils, an interstage oxygen feed, and separate irradiation modules; the amidation sequence used two heated ETFE coils and interstage benzylamine addition. Both used sample-loop injection and a 7 bar cartridge BPR. NMR yields were determined using 1,3-benzodioxole as the internal standard; isolated products were characterized by  $^1\text{H}/^{13}\text{C}$  NMR and LRMS. ESI Section 12 contains the supplied apparatus photographs, procedures, numerical reconciliation, and outstanding metrology details. **Figures S28-S31** contain the product-characterization spectra. No replicate SD or measured two-phase residence time is inferred where these records were not supplied.

## AUTHOR INFORMATION

### *Corresponding Authors*

\*Boyoung Y. Park, E-mail: boyoungy.park@khu.ac.kr; \*Gwang-Noh Ahn, E-mail: gnahn@kriect.re.kr

### *Author Contributions*

A.M. and M.P. contributed equally to this work. The complete contributor-role statement and approval of the final author list will be confirmed by all authors before submission.

## ***Notes***

### **Availability of data and code**

Source code is publicly accessible at <https://github.com/armtfgh/flowpilot>. Benchmark scripts and the data-access index are available at [https://github.com/armtfgh/flowpilot/tree/master/ablation\\_test](https://github.com/armtfgh/flowpilot/tree/master/ablation_test) and [https://github.com/armtfgh/flowpilot/tree/master/ablation\\_results](https://github.com/armtfgh/flowpilot/tree/master/ablation_results), respectively. The latter is currently an index, not the complete outcome archive. The ESI contains the reported numerical summaries, experimental procedures, and characterization spectra. As of 29 September 2026, the repository is public but has no tagged manuscript release; the complete benchmark archive, raw experimental analytical files, and a persistent data DOI have not yet been deposited. These release items remain to be completed before submission.

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## **ABBREVIATIONS**

BPR, back-pressure regulator; DPDTC, dipyridyldithiocarbonate; DMAP, 4-(dimethylamino)pyridine; Da, Damköhler number; ETFE, ethylene tetrafluoroethylene; FEP, fluorinated ethylene propylene; IF, intensification factor; LLM, large language model; LRMS, low-resolution mass spectrometry; MFC, mass flow controller; 2-MeTHF, 2-methyltetrahydrofuran; NMR, nuclear magnetic resonance; Pe, Péclet number; PFA, perfluoroalkoxy; PFD, process flow diagram; PTFE, polytetrafluoroethylene; RAG, retrieval-augmented generation; Re, Reynolds number; STP, standard temperature and pressure (273.15 K and 1 atm for the proposal calculations); STY, space-time yield.

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